```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LinearRegression, Ridge, Lasso

from sklearn.metrics import r2_score, mean_squared_error

df = pd.read_csv("uber.csv")

df .head()

df.isnull().sum()

df["pickup_datetime"] = pd.to_datetime(df["pickup_datetime” ])

df.dropna(inplace=True)

if "Unnamed: @" in df.columns:
df.drop("Unnamed: @", axis=1, inplace=True)

df.describe()

plt.figure(figsize=(8, 5))
plt.boxplot(df["fare_amount"])

plt.title("Boxplot of Fare Amount")
plt.ylabel("Fare Amount")

plt.show()

q1= df["fare_amount"].quantile(@.25)
q3= df["fare_amount"].quantile(@.75)

igr = q3 - ql
```

```
lb = qi - 1.5 * iqr
ub = q3 + 1.5 * iqr
df = df[(df["fare_amount"] >= 1b) & (df["fare_amount"] <= ub) ]

plt.boxplot(df["fare_amount" ])
plt.title("Fare Amount after Removing Outliers")
plt.show()

numeric_cols = df.select_dtypes(include=np.number )
correlation_matrix = numeric_cols.corr()

plt.figure(figsize=(8,6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm' )

plt.title("Correlation Matrix")
plt.show()

df[["pickup_longitude", "pickup_latitude", "dropoff_longitude", "“dropoff_latitu
df["fare_amount"]]

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42

scaler = StandardScaler()
X_train_scaled = scaler. fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)

y_pred_lr = lr_model.predict(X_test_scaled)

r2_lr = r2_score(y_test, y_pred_lr)
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_l1r))

print("Linear Regression -> R2 Score:", r2_l1r)
print("Linear Regression -> RMSE:", rmse_l1r)

ridge model = Ridge(alpha=1.0)
ridge model.fit(X_train_scaled, y train)

y_pred_ridge = ridge _model.predict(X_test_scaled)
```

```
r2_ridge = r2_score(y_test, y_pred_ridge)
rmse_ridge = np.sqrt(mean_squared_error(y_test, y_pred_ridge) )

print("Ridge Regression -> R2 Score:", r2_ridge)
print("Ridge Regression -> RMSE:", rmse_ridge)

Ridge Regression -> R2 Score: @.0004036479501785539
Ridge Regression -> RMSE: 3.6373665936303716

lasso_model = Lasso(alpha=0@.1)
lasso_model.fit(X_train_scaled, y_train)

y_pred_lasso = lasso_model.predict(X_test_scaled)

r2_lasso = r2_score(y_test, y_pred_lasso)
rmse_lasso = np.sqrt(mean_squared_error(y_test, y_pred_lasso))

print("Lasso Regression -> R2 Score:", r2_lasso)
print("Lasso Regression -> RMSE:", rmse_lasso)

Lasso Regression -> R2 Score: -3.106368247585145e-05
Lasso Regression -> RMSE: 3.638157429697902

results = pd.DataFrame({
"Model": ["Linear Regression", "Ridge Regression", "Lasso Regression" ],
"R2 Score": [r2_lr, r2_ridge, r2_lasso],
"RMSE": [rmse_lr, rmse_ridge, rmse_lasso]

})

print(results)
```